

Dataset Analysis for Decision-Making

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 sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and address your request, I will first load the dataset (`sicor_municipal_contracts.csv`) and examine its structure and contents. This will help identify key characteristics such as column names, data types, missing values, and general trends. Once the dataset's structure is understood, I will be ready to address any specific analytical question you pose.

Below is the Python code to load and inspect the dataset:

Step 1: Load and Inspect the Dataset

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
14
15 # Check for missing values
16 print("\nMissing Values in Each Column:")
17 print(data.isnull().sum())
```

Step 2: Key Characteristics of the Dataset

After running the above code, I will summarize:

1. **Column Names** : What are the variables included in the dataset?
2. **Data Types** : Are the columns numeric, categorical, or date/time?
3. **Missing Data** : Are there any missing values that need to be addressed?
4. **General Trends** : Initial observations about the distribution of values or notable patterns.

Step 3: Address Your Analytical Question

Once you provide a specific analytical question, I will write Python code to extract and process the relevant data, followed by a clear and concise answer.

Please provide your analytical question so I can proceed with the analysis.

How does the subprogram classification influence the total contract value?

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To address the question of how the subprogram classification influences the total contract value, we need to analyze the dataset by grouping the data based on the `subprogram` column (or a similar column that contains subprogram



classifications) and then calculating the total contract value for each subprogram.

Below is the Python code to perform this analysis:

Step 1: Analyze Subprogram Influence on Total Contract Value

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sior_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Check if 'subprogram' and 'contract_value' columns exist
8 if 'subprogram' not in data.columns or 'contract_value' not in data.columns:
9     raise ValueError("The dataset must contain 'subprogram' and 'contract_value' columns.")
10
11 # Group by subprogram and calculate the total contract value for each subprogram
12 subprogram_totals = data.groupby('subprogram')['contract_value'].sum().reset_index()
13
14 # Sort the results by total contract value in descending order
15 subprogram_totals = subprogram_totals.sort_values(by='contract_value', ascending=False)
16
17 # Display the top subprograms by total contract value
18 print("\nTop Subprograms by Total Contract Value:")
19 print(subprogram_totals.head())
20
21 # Optionally, visualize the results using a bar plot
22 import matplotlib.pyplot as plt
23
24 plt.figure(figsize=(10, 6))
25 plt.barh(subprogram_totals['subprogram'].head(10), subprogram_totals['contract_value'].head(10),
26          color='blue')
27 plt.xlabel('Total Contract Value')
28 plt.ylabel('Subprogram')
29 plt.title('Top 10 Subprograms by Total Contract Value')
30 plt.gca().invert_yaxis() # Invert y-axis for better readability
31 plt.show()
```

Step 2: Explanation of the Code

- 1. **Column Verification** : The code first checks if the dataset contains the necessary columns (`subprogram` and `contract_value`). If these columns are missing, an error is raised.
- 2. **Grouping and Aggregation** : The dataset is grouped by the `subprogram` column, and the total contract value for each subprogram is calculated using the `.sum()` function.
- 3. **Sorting** : The results are sorted in descending order of total contract value to identify the subprograms with the highest values.
- 4. **Visualization** : A horizontal bar plot is generated to visually represent the top subprograms by total contract value.

Step 3: Deliverables

- 1. **Answer** : After running the code, the output will provide a ranked list of subprograms based on their total contract value. This will help identify which subprograms have the most significant financial impact.
- 2. **Visualization** : The bar plot will highlight the top subprograms, making it easier to interpret the results.

Example Output (Hypothetical)

If the dataset contains the following data:

SUBPROGRAM	CONTRACT_VALUE
Infrastructure	5000000
Education	3000000
Healthcare	7000000
Environment	2000000

The output might look like this:

```
1 Top Subprograms by Total Contract Value:
2   subprogram  contract_value
3   -         -
4   Infrastructure  5000000
5   Education      3000000
6   Healthcare      7000000
7   Environment     2000000
```

3	0	Healthcare	7000000
4	1	Infrastructure	5000000
5	2	Education	3000000
6	3	Environment	2000000

And the bar plot would show "Healthcare" at the top, followed by "Infrastructure," "Education," and "Environment."

Please confirm if the column names (`subprogram` and `contract_value`) match your dataset, or provide the correct column names so I can adjust the analysis accordingly.

